Tariff Switch Processing System – Case Study Specification

Overview

This case study describes a lightweight .NET console application written in C# that automates the processing of customer tariff-switch requests. The application reads structured data from CSV files and applies validation rules to decide whether to approve or reject each request. If a request is valid, the specified tariff should be applied to the customer before the SLA due date. If not, the request must be rejected with a clear reason.

Core Goals

- Validation of input data – ensure that requests reference valid customers and tariffs and that customers have no unpaid invoices.
- Business logic enforcement – handle conditions such as smart meter requirements and SLA-based deadlines.
- Idempotent processing – ensure that each request is processed only once, even across multiple runs.
- Time-zone-aware SLA computation – compute deadlines in Europe/Vienna with correct daylight-saving time handling.

Input Files

- customers.csv – Customer master data (ID, Name, Unpaid-invoice flag, SLA level, Meter type).
- tariffs.csv – Tariff catalog (ID, Name, Smart-meter requirement, Monthly price).
- requests.csv – Tariff-switch requests (Request ID, Customer ID, Target Tariff ID, Requested timestamp).

Approval and Rejection Scenarios (Business Logic)

1. Approve – Standard SLA, no unpaid invoices, no smart meter required or already installed. SLA due = RequestedAt + 48h (Europe/Vienna).
2. Approve – Premium SLA, no unpaid invoices, no smart meter required or already installed. SLA due = RequestedAt + 24h (Europe/Vienna).
3. Approve – Tariff requires a smart meter, and the customer has a classic meter. Add follow-up action 'Schedule meter upgrade'. SLA due = (Standard/Premium rule) + 12h.
4. Reject – Customer has unpaid invoices. Reason: 'Unpaid invoice'.
5. Reject – Unknown customer ID. Reason: 'Unknown customer'.
6. Reject – Unknown tariff ID. Reason: 'Unknown tariff'.
7. Reject – Invalid request data (bad timestamp or malformed fields). Reason: 'Invalid request data'.
8. Skip – Already processed in a previous run. Do not process again.

Date Time Rules

- Compute all SLA due times using Europe/Vienna local time.
- Ensure DST-safe calculations: durations must follow local time semantics.
- Output timestamps using ISO-8601 format.

Expected Application Behavior

On each run, the application loads the three CSV input files and processes only the requests that have not yet been handled. Previously processed requests must not be processed again. When new rows are added to requests.csv, subsequent runs should process only the new entries. Row-level data problems (such as unknown customer IDs, unknown tariff IDs, or invalid timestamps) should cause the request to be rejected with a clear reason, while the rest of the processing continues. However, configuration or schema errors (for example missing files or missing columns) must cause the application to fail fast with a clear error message so the user can correct the input. The application must always use the most recent data from the CSV files and seamlessly handle newly added customers, tariffs, and requests.

Deliverables

The final solution should include the application source code, instructions for running the application, and automated tests verifying the core logic. The solution can be delivered as a repository containing the complete project structure.